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EX NO: 8a	SIMULATION AND ANALYSIS OF BPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB
DATE:	

Objective

- i) To generate BPSK symbols from a binary data sequence using phase-shift keying principles.
- ii) To Plot and analyze the BPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.
- iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation in symbol detection performance.

Matlab code

1	clc;	/MATLAB Drive/experiment_7.m	31	xlabel('In-Phase (I)');
2	clear;		32	ylabel('Quadrature (Q)');
3	close all;		33	title('Ideal BPSK Constellation');
4			34	
5	%% Objective 1: Generate BPSK symbols		35	%% Objective 3: Add Noise and Evaluate Performance
6	N = 1000; % Number of bits		36	SNR = 5; % Signal-to-Noise Ratio (dB)
7	data = randi([0 1], N, 1); % Random binary sequence		37	
8			38	rx_signal = awgn(bpsk_symbols, SNR, 'measured');
9	% BPSK Mapping		39	
10	bpsk_symbols = 2*data - 1;		40	subplot(1,2,2);
11			41	
12	%% Objective 2: Plot BPSK Signal Constellation		42	scatter(real(rx_signal), imag(rx_signal), ...
13	figure;		43	15, 'filled');
14			44	
15	subplot(1,2,1);		45	hold on;
16			46	
17	scatter(real(bpsk_symbols), imag(bpsk_symbols), ...		47	xline(0, 'k', 'LineWidth', 1.5);
18	30, 'filled');		48	yline(0, 'k', 'LineWidth', 1.5);
19			49	
20	hold on;		50	grid on;
21			51	axis equal;
22	xline(0, 'k', 'LineWidth', 1.5);		52	
23	yline(0, 'k', 'LineWidth', 1.5);		53	xlim([-2.5 2.5]);
24			54	ylim([-2 2]);
25	grid on;		55	
26	axis equal;		56	xlabel('In-Phase (I)');
27			57	ylabel('Quadrature (Q)');
28	xlim([-1.5 1.5]);		58	
29	ylim([-1 1]);		59	title(['Noisy BPSK Constellation (SNR = ', ...
	/MATLAB Drive/experiment_7.m			
55				
56	xlabel('In-Phase (I)');			
57	ylabel('Quadrature (Q)');			
58				
59	title(['Noisy BPSK Constellation (SNR = ', ...			
60	num2str(SNR), ' dB']);			
61				
62	%% Symbol Detection			
63	detected_bits = real(rx_signal) > 0;			
64				
65	%% Calculate Errors			
66	num_errors = sum(data ~= detected_bits);			
67	BER = num_errors / N;			
68				
69	%% Display Results			
70	fprintf('Number of Transmitted Bits : %d\n', N);			
71	fprintf('Number of Bit Errors : %d\n', num_errors);			
72	fprintf('Bit Error Rate (BER) : %f\n', BER);			

Procedure:

Objective 1: Generate BPSK Symbols from a Binary Data Sequence Using Phase-Shift

Keying Principles

1. Initialize the MATLAB environment and define the number of bits.
2. Generate a random binary sequence consisting of 0s and 1s.
3. Apply BPSK modulation by mapping binary 0 to -1 and binary 1 to $+1$.
4. Store the generated BPSK symbols for further processing.
5. Verify that the symbol sequence contains only two distinct symbol values.
6. Prepare the modulated symbols for constellation analysis.

Model Graph

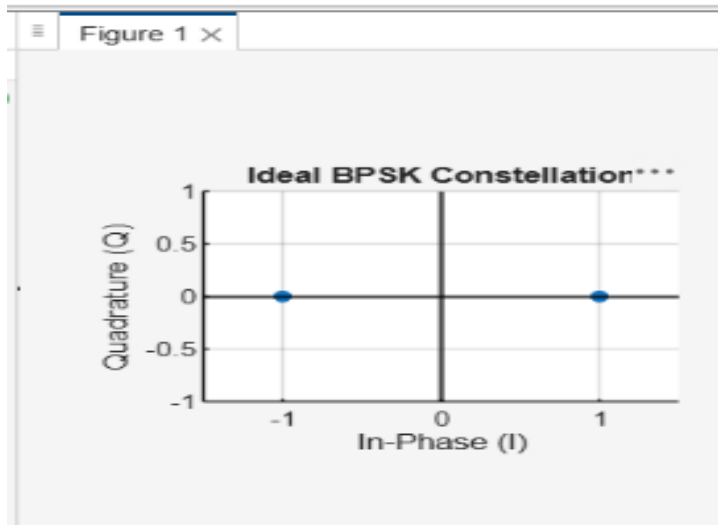


Fig: Signal Constellation for BPSK Symbols

Result BPSK symbols were successfully generated from the binary data sequence using phase-shift keying modulation.

Objective 2: Plot and Analyze the BPSK Signal Constellation in the In-Phase (I) and Quadrature (Q) Plane

1. Use the generated BPSK symbols as inputs for constellation plotting.
2. Assign the real part of the symbols to the In-Phase (I) axis.
3. Set the Quadrature (Q) component to zero for all symbols.
4. Plot the constellation points on the I-Q plane using a scatter plot.
5. Draw the horizontal and vertical axes through the origin and label them appropriately.
6. Analyze the constellation diagram and identify the symbol locations corresponding to binary 0 and binary 1.

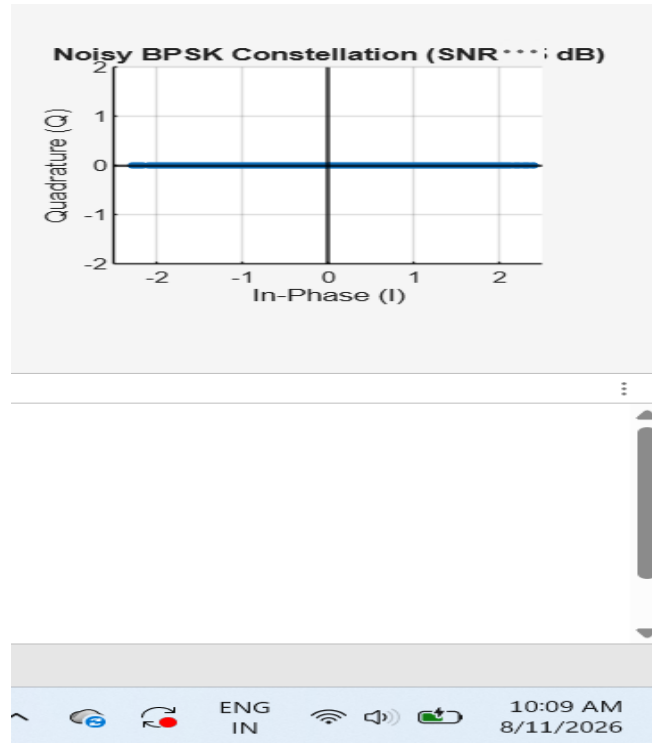


Fig. 2: Noisy BPSK Constellation

Result : The BPSK constellation was successfully plotted in the In-Phase (I) and Quadrature (Q) plane, showing two distinct symbol points corresponding to binary 0 and binary 1 for the given SNR.